

# Manish

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## Skills

**Languages:** Python, JavaScript, SQL

**Frameworks & Libraries:** ReactJS, FastAPI, SQLAlchemy, TailwindCSS, PyTorch, Hugging Face, LangChain

**Core Areas:** Full-Stack Web Development, RESTful API Design, WebSocket, WebRTC, Authentication & Authorization (basic), Natural Language Processing, Generative AI, Retrieval-Augmented Generation, Fine-tuning

**Tools & Platforms:** Git, GitHub, Docker, PostgreSQL, Firebase, Neo4j, Ollama, MLflow

## Education

**Kings Engineering College - Chennai**

Sep 2022 - Apr 2026

B.Tech. in Artificial Intelligence and Data Science

**CGPA: 8.3/10**

Relevant Coursework: Object Oriented Programming, Databases, Discrete Maths, Statistics and Probability, Data Structures and Algorithms, Computer Networks, Exploratory Data Analysis, Machine Learning, Deep Learning, Big Data

## Project Work

- **Real-Time Chat Application:**
  - Built a full-stack messaging application supporting one-to-one and group chats, voice/video calls, and 24-hour status stories, similar in scope to WhatsApp.
  - Implemented a single persistent WebSocket connection to handle typing indicators, online presence, read receipts, and WebRTC call signaling for voice and video calls.
  - Designed JWT-based authentication with refresh token rotation, role-based group membership, and granular per-user privacy controls.
  - Technologies: FastAPI, SQLAlchemy, PostgreSQL, WebSocket, WebRTC, React, TailwindCSS, Axios.
- **KM Gurukulam – School Website (Live):**
  - Collaborated with a teammate to design and build a live website for a toddler school, currently deployed and used by the client for admissions.
  - Styled a fully responsive interface across the landing page, admissions form, and gallery sections, and handled the overall UI polish and layout.
  - Set up Firebase and Supabase as backend services to store admission submissions, and built an admin dashboard to search, filter, and export applicant data to Excel.
  - Technologies: Next.js, Firebase, Supabase, TailwindCSS, Radix UI.
- **AI Avatar Assistant System:**
  - Built the full-stack architecture for an AI avatar assistant, connecting a React frontend to an async FastAPI backend through JWT-secured REST APIs.
  - Designed the authentication system with access/refresh tokens and session handling, and structured the PostgreSQL schema using SQLAlchemy's async ORM with Alembic migrations.
  - Built a background task scheduler for automatic media cleanup and orchestrated a multi-stage processing pipeline (speech-to-text, LLM response, text-to-speech, lip-sync) behind a single API layer.
  - Technologies: FastAPI, SQLAlchemy, PostgreSQL, JWT, React, Axios, Vite.
- **StudyGraphRAG:**
  - Built a full-stack study assistant with a FastAPI backend and React frontend that lets users upload PDF documents and query them through a Neo4j knowledge graph.
  - Designed asynchronous background workers to parse documents, extract entities and relationships, and generate embeddings without blocking the main API.
  - Built REST APIs for document upload, graph management, and conversational querying, combining retrieved graph context with an LLM to generate answers.
  - Technologies: FastAPI, SQLAlchemy, React, TailwindCSS, Axios, Neo4j, GraphRAG, Gemini API.

## Awards and Certificates

- **Vice President, AI Club (College):** Organized AI and machine learning seminars, mentored junior students, and conducted hands-on technical sessions.
- **Chatbot Challenge Runner-Up:** Secured 2nd place in a chatbot development competition at Sairam Institute by building a functional chatbot within 60 minutes.
- Completed Machine Learning and Data Science courses on Udemy covering supervised learning, deep learning, NLP, and basics of model deployment.